Baseline Models Report

Economic Forecasting for Banking Using FRED Time Series: Phase 1

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Background & Question

Research question. Can a leakage-controlled set of exogenous FRED macro series support chronological forecasts of the weekly 30-year fixed mortgage rate for bank risk and treasury use, and under what conditions does that signal weaken?

Hypothesis. Treasury yields and policy rates carry usable predictive information. An exogenous-only model should beat simple linear baselines on a post-2021 hold-out, while larger errors should appear when the mortgage–Treasury spread widens under rapid tightening.

Methods

After feature engineering, the plan was to fit a small set of baselines on the same clean exogenous matrix, then compare them to a tree model.

Baselines chosen. Random walk is the classical persistence floor for interest rates. OLS is the linear exogenous benchmark. SARIMAX(1,1,1) with exogenous regressors is the classical time-series plus macros baseline. Bayesian Ridge places a Gaussian prior on coefficients and returns predictive bands. LassoCV is the sparse regularized linear model. LightGBM (exogenous-only) is the non-linear improvement candidate.

All models use the same chronological 80/20 split (train through early 2021; test March 2021 through mid-2026) and the same rule: no features derived from the mortgage rate itself.

Cross-validation and complexity control. Lasso used 5-fold cross-validation inside the training window only to choose the penalty. Bayesian Ridge shrinks coefficients via its prior. LightGBM used shallow trees (depth 4, 15 leaves). Final metrics are always evaluated on the chronological test set, never on reshuffled data.

Assumptions tested. OLS, Bayesian Ridge, and Lasso assume approximate linearity and Gaussian-like noise. These were checked with residual plots, Durbin–Watson statistics, residual

ACF/PACF, and Ljung–Box tests. Bayesian Ridge also reports approximate 90% posterior band coverage. LightGBM does not assume Gaussian errors; residual structure and SHAP values are used instead.

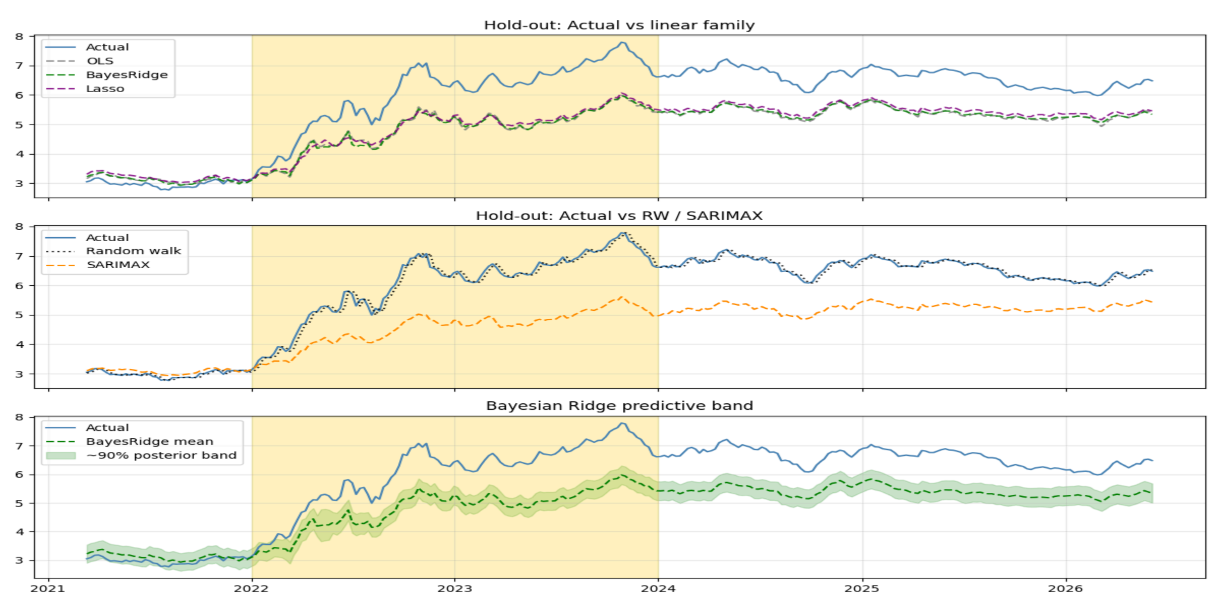
Results

(1) Fitted model

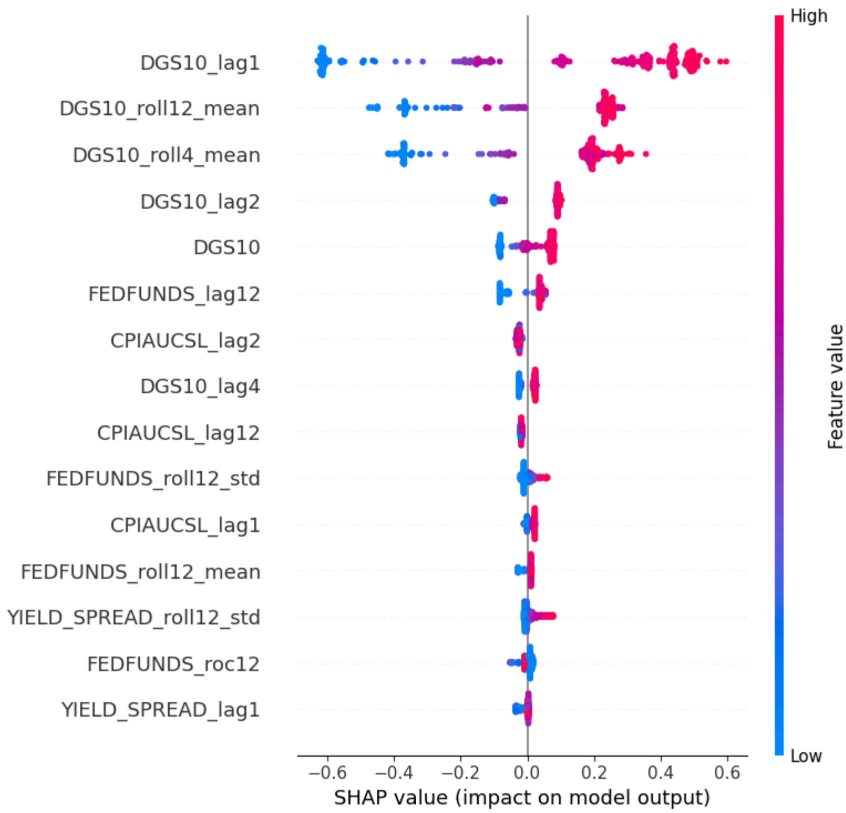
Hold-out performance on the chronological test set is summarized below.

Model	Test R^2	Mean resid	2022–23 mean resid
Random walk	0.993	0.01	0.03
LightGBM (exog-only)	0.833	0.42	0.73
LightGBM + regime interactions	0.844	0.40	0.72
LassoCV	0.421	0.93	1.20
Bayesian Ridge	0.351	1.01	1.25
OLS	0.344	1.01	1.24
SARIMAX(1,1,1)+exog	0.105	1.18	1.51

The random walk wins pure one-step persistence. Among exogenous models, LightGBM roughly doubles Lasso's R-squared and cuts average under-prediction in half. Linear and Bayesian linear paths all sit well below actual rates through 2022–2023. Lasso is the best linear baseline; SARIMAX is the weakest.



SHAP analysis on LightGBM is dominated by the one-week lag of the 10-year Treasury yield and Treasury rolling means. Effects are monotone and economically sensible: higher lag-1 Treasury values raise the predicted mortgage rate. Housing starts rank low; removing the HOUST feature family does not reduce test R-squared.



(2) Assumptions

Durbin–Watson statistics are approximately 0 for OLS, Lasso, and Bayesian Ridge, and about 0.03 for LightGBM. Residuals are highly positively autocorrelated. Residual ACF remains above about 0.9 at lag 12 for linear models and about 0.72 for LightGBM. Partial autocorrelation spikes at lag 1 and then dies out, which is consistent with level bias rather than a rich residual autoregressive structure. Ljung–Box tests reject white-noise residuals for all structural models (p near 0).

Bayesian Ridge approximate 90% posterior band coverage is only 0.18 overall and 0.07 in 2022–2023. Bands are centered on a biased mean and therefore miss the actual path. Gaussian linear assumptions do not hold under the tightening regime; that is reported as a finding rather than assumed away.

(3) Overfitting

Overfitting is not the main issue. Controls in place include a chronological hold-out, an exogenous-only feature matrix (prior ablation showed that mortgage lags inflate hold-out R-squared to about 0.98 and were removed), Lasso sparsity (15 of 61 features), Bayesian coefficient shrinkage, and shallow trees. The dominant failure mode is regime misspecification—the mortgage–Treasury spread widening—not memorization of the training sample.

Discussion & Next Steps

Takeaways. Exogenous macros do carry signal: LightGBM recovers substantial chronological skill (R-squared about 0.83) versus linear baselines (about 0.35–0.42). The hypothesis is supported on that point. It is also supported on the risk side: errors concentrate when spreads widen. Residual autocorrelation and SHAP show that the model already relies on Treasuries; the leftover bias is the mortgage–Treasury disconnect under rapid tightening, not omitted housing starts or classical overfitting.

Analysis plan revisions. Keep the random walk as the persistence floor (not a model to beat at one-week-ahead). Treat Lasso as the interpretable linear baseline and LightGBM as the primary structural model. Regime interactions help only slightly and should not be treated as a full fix. Monte Carlo and multi-step evaluation remain secondary until the one-step exogenous fit is locked.

Next steps;

Hyperparameter Tuning and Model Evaluation:

1. Additional models: locked LightGBM as primary; optional Elastic Net; no reintroduction of target lags.

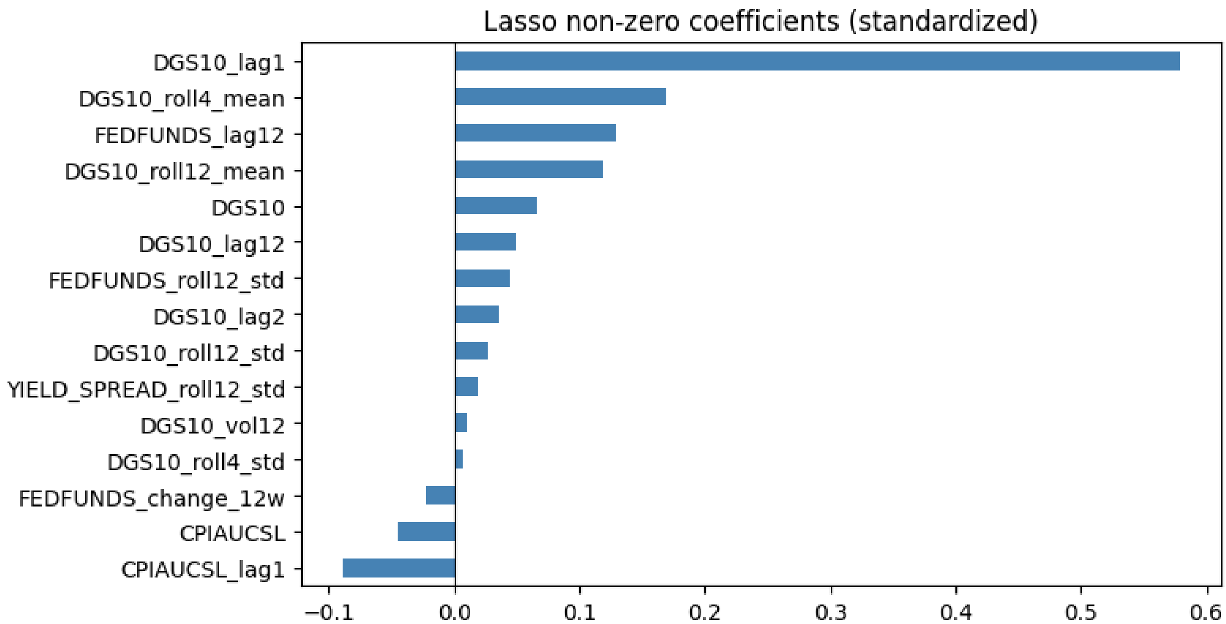
2. Tuning: time-ordered or expanding-window search inside the training sample only for learning rate, depth, and number of leaves.

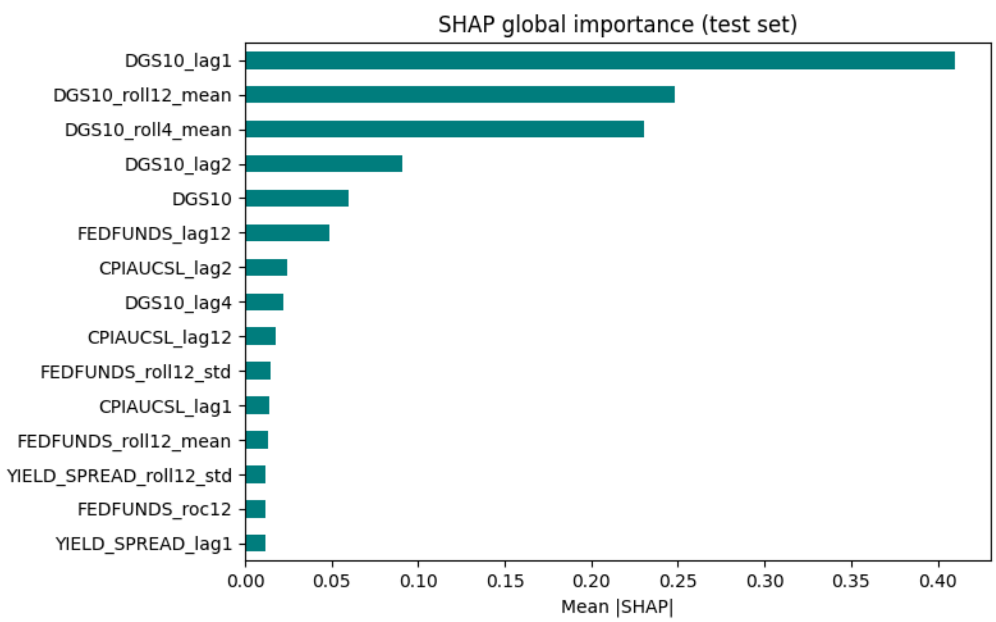
3. Validation: same chronological test; report MSE, R-squared, mean residual, and 2022–2023 residual behavior; residual ACF as a diagnostic, not a fit target.

4. Plan emphasis: exogenous signal with documented regime limits, rather than multi-step risk metrics, until Phase 2.

Appendix — Data Dictionary and Supplemental Plots

Variable / family	Type	Description
MORTGAGE30US	Target	30-year fixed mortgage rate (weekly)
DGS10, FEDFUNDS, UNRATE, CPIAUCSL, HOUST	Exogenous levels	Treasury, policy, labor, inflation, housing starts
YIELD_SPREAD	Engineered	DGS10 minus FEDFUNDS
*_lag1,2,4,8,12	Engineered	Lags of exogenous series only
*_roll4/12_mean/std	Engineered	Rolling mean/std of selected exogenous series
_roc	Engineered	Rate of change of selected exogenous series
FEDFUNDS_change_12w, tightening_flag, DGS10_vol12	Regime	Policy impulse, binary stress flag, Treasury volatility


Lasso non-zero coefficients (standardized)
DGS10_lag1
DGS10_roll4_mean
FEDFUNDS_lag12
DGS10_roll12_mean
DGS10
DGS10_lag12
FEDFUNDS_roll12_std
DGS10_lag2
DGS10_roll12_std
YIELD_SPREAD_roll12_std
DGS10_vol12
DGS10_roll4_std
FEDFUNDS_change_12w
CPIAUCSL
CPIAUCSL_lag1
−0.1
0.0
0.1
0.2
0.3
0.4
0.5
0.6


SHAP global importance (test set)
DGS10_lag1
DGS10_roll12_mean
DGS10_roll4_mean
DGS10_lag2
DGS10
FEDFUNDS_lag12
CPIAUCSL_lag2
DGS10_lag4
CPIAUCSL_lag12
FEDFUNDS_roll12_std
CPIAUCSL_lag1
FEDFUNDS_roll12_mean
YIELD_SPREAD_roll12_std
FEDFUNDS_roc12
YIELD_SPREAD_lag1
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
0.40
Mean |SHAP|